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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

EE2252-1 – HYBRID ELECTRIC VEHICLES

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is a benefit of the series configuration in electric vehicles?
A) Simplicity of design
B) Improved fuel efficiency under all conditions
C) Direct power from the engine to the wheels
D) High performance in all driving conditions
2. What can significantly increase energy consumption in an electric vehicle?
A) Driving at lower speeds
B) Light cargo
C) Using air conditioning
D) Regenerative braking
3. The efficiency of an electric vehicle can be adversely affected by:
A) High speeds
B) Aerodynamic design
C) Optimal tire pressure
D) Low temperatures
4. Which standard is critical for ensuring the functional safety of automotive systems?
A) ISO 9001
B) ISO 14001
C) ISO 26262
D) ISO 50001
5. Which of the following describes the operational strategy of hybrid vehicles?
A) Solely electric or gasoline
B) Relying only on fossil fuels
C) Dual energy sources for optimal efficiency
D) Solar-powered
6. What distinguishes a parallel hybrid electric drive train from a series hybrid?
A) Engine and motor can work independently or together
B) It only uses an electric motor
C) It has a more complex electrical system
D) It operates only in electric mode
7. How does regenerative braking function in parallel hybrids?
A) Only the engine captures energy
B) It has no regenerative capabilities
C) The electric motor converts kinetic energy back into electrical energy
D) The brakes are always mechanical
8. What type of battery is most commonly used in hybrid electric vehicles?
A) Lead-acid
B) Nickel-cadmium
C) Lithium-ion
D) Alkaline
9. What primarily causes grading resistance?
A) Vehicle weight
B) Tire pressure
C) Incline or decline of the road
D) Wind speed
10. How does aerodynamic drag affect vehicle performance?
A) Decreases fuel efficiency
B) Opposes forward motion at high speeds
C) Increases speed
D) Has no significant effect
11. What does the thermostat control strategy do in hybrid vehicles?
A) Turns the engine on or off based on temperature
B) Manages fuel injection
C) Controls tire pressure
D) Adjusts aerodynamic features

12. What is a key design principle of series hybrid systems?
 A) Sole reliance on fossil fuels B) Use of electric motors for propulsion
 C) Elimination of batteries D) Direct mechanical connection between engine and wheels
13. What is the primary characteristic of a parallel hybrid electric drive train?
 A) Only the electric motor drives the wheels B) Both the internal combustion engine and electric motor can drive the wheels
 C) It uses only one type of power source D) It is always powered by the battery alone
14. What is meant by parametric design in the context of hybrid drive trains?
 A) Simplifying the drive train layout B) Selection of colors and materials
 C) Designing components based on specific performance parameters D) Aesthetic design of the vehicle
15. Which parameter is crucial when sizing the electric motor in a hybrid drive train?
 A) Color of the vehicle B) Tire design
 C) Required power output for acceleration D) Vehicle weight
16. Which factor is typically NOT considered in the parametric design of a drive train?
 A) Driver behavior B) Brand loyalty
 C) Terrain type D) Vehicle speed
17. What is a key consideration in the design of energy storage for PHEVs?
 A) Engine power output B) Aesthetic design
 C) Weight of the vehicle D) Battery capacity and charging efficiency
18. What is a primary function of the CAN protocol?
 A) To manage tire pressure B) To enable communication between vehicle components
 C) To control the engine size D) To reduce vehicle speed
19. Which of the following is a feature of CAN message frames?
 A) Limited to 8 bits B) Variable size with identifier and data field
 C) Non-priority based D) Fixed size only
20. What is the maximum data rate supported by CAN?
 A) 100 Mbps B) 10 Mbps
 C) 1 Mbps D) 1 kbps

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1.	a) Explain how the tractive effort developed by a traction motor on driven wheels and the vehicle speed can be calculated.	6	L*2	1	1,2
	b) With the neat diagram explain the parallel hybrid electric drive train and outline its main components.	4	L2	2	1,2
	c) Explain with neat curves the Power rating versus speed factor of electric vehicles.	6	L2	1	1,2
2.	a) Explain with the diagram the mechanical torque coupler for parallel hybrid electric drive train.	6	L2	2	1,2
	b) Explain with neat diagram the Tractive effort versus vehicle speed characteristics for traction motor of $x = 2$ and 3-gear transmission.	4	L2	1	1,2
	c) With the neat diagram explain the Configuration of a series hybrid electric drivetrain.	6	L2	2	1,2
3.	a) With the neat diagram explain the separated axle torque combination parallel hybrid electric drivetrain	6	L2	2	1,2
	b) Explain the neat the mechanical speed coupler for parallel hybrid drive train.	4	L2	2	1,2
	c) Explain with neat diagram the Acceleration time and distance versus final speed curve.	6	L2	1	1,2

Unit – II

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| 4. | a) | Explain what is meant by Aerodynamic drag? Explain its components: | 6 | L2 | 3 | 1,2 |
| | b) | With the neat diagram explain the overall control scheme of parallel torque-coupling hybrid drivetrain. | 4 | L2 | 4 | 1,2 |
| | c) | Explain the mathematical relationship used to calculate grading resistance based on a vehicle's weight and the angle of the incline. | 6 | L3 | 3 | 1,2 |
| 5. | a) | With the neat diagram explain the engine characteristics and optimal operating region. | 6 | L2 | 3 | 1,2 |
| | b) | Explain the tractive effort versus vehicle speed characteristics of electric-motor-driven vehicle of parallel hybrid electric vehicle. | 6 | L2 | 4 | 1,2 |
| | c) | Explain with the neat diagram the engine on-off control strategy for parallel hybrid electric drive train. | 4 | L2 | 4 | 1,2 |
| 6. | a) | Explain with equations how to calculate the engine power capacity in parallel hybrid vehicle. | 6 | L2 | 4 | 1,2 |
| | b) | Explain with neat diagram the maximum State of Charge (SOC) control strategy for hybrid vehicles. | 6 | L2 | 3 | 1,2 |
| | c) | With the neat diagram explain the configuration of a typical series hybrid electric drivetrain | 4 | L2 | 3 | 1,2 |

Unit – III

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|----|----|---|---|----|---|-----|
| 7. | a) | Describe the fundamental principles of the Controller Area Network (CAN). | 6 | L2 | 5 | 1,2 |
| | b) | Explain AER focused energy management strategies in hybrid vehicles? | 4 | L2 | 5 | 1,2 |
| | c) | Explain with the neat diagram the Daily driving distance distribution and cumulative factor. | 6 | L2 | 5 | 1,2 |
| 8. | a) | How does CAN facilitate communication between various vehicle systems? | 6 | L2 | 5 | 1,2 |
| | b) | Explain how to calculate the total energy capacity of the energy storage system. | 6 | L2 | 5 | 1,2 |
| | c) | Explain how ideal torque-speed characteristics and maximum motoring power of motor can be expressed by equations. | 4 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
